

Energy Ledgers for Forced Harmonic ODEs

Kirchhoff power balance and an unknown component with $X = \text{LRC}$

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Abstract

An added term Xq''' supplies or absorbs energy according to the sign of $Xq'''q'$. Exact trajectories show both roles. Choosing $X = \text{LRC}$ scales the transfer on fixed motion; it does not choose its direction. Hammer and rotary examples apply the same rule and give apparent coefficients of performance relative to specified ordinary inputs. Their counted output includes resistance work and elastic storage; the component's physical identity and remaining energy account stay open.

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1 From voltage balance to power accounts

Start with the forced [LRC template](#), Section 2:

$$Lq'' + Rq' + \frac{q}{C} = f(t), \quad L, C > 0, \quad R \geq 0. \quad (1)$$

Here q is charge, $i = q'$ current, f forcing voltage, and L, R, C inductance, resistance and capacitance. Coefficients are constant, trajectories smooth, and primes denote time derivatives. In the [series convention](#), Section 1, the same current multiplies every voltage term. Put the source first and record the circuit power balance in line 2, with opposite transfers in lines 1 and 3:

$$\begin{cases} 1 : & \dots + Rq'i + \dots = 0, \\ 2 : & f(t)i - Lq''i - Rq'i - qi/C = 0, \\ 3 : & -f(t)i + \dots = 0. \end{cases} \quad (2)$$

Line 1 belongs to the heat recipient and line 3 to the source or sink; $fi < 0$ means the source receives energy. Opposite entries record each transfer on both sides. Constant total energy also requires the omitted energy rates and reservoir laws; prescribed $f(t)$ alone gives no source capacity.

2 When the added term supplies or absorbs

Add a fictional component with constant $X > 0$ and unknown physical identity:

$$Xq''' + Lq'' + Rq' + \frac{q}{C} = f(t). \quad (3)$$

Treating Xq''' as its voltage contribution adds a fourth account:

$$\begin{cases} 1 : & \cdots + Rq'i + \cdots = 0, \\ 2 : & f(t)i - Xq'''i - Lq''i - Rq'i - qi/C = 0, \\ 3 : & -f(t)i + \cdots = 0, \\ 4 : & \cdots + Xq'''i + \cdots = 0. \end{cases} \quad (4)$$

Thus the received power and the contribution to line 2 are

$$P_X = Xq'''i, \quad P_{X \rightarrow 2} = -Xq'''i. \quad (5)$$

Since $X > 0$, the component supplies power when $q'''i < 0$, absorbs it when $q'''i > 0$, and transfers none when the product is zero. Over an interval,

$$W_X(t_0, t_1) = \int_{t_0}^{t_1} Xq'''(t)i(t) dt, \quad W_{X \rightarrow 2} = -W_X. \quad (6)$$

Positive W_X is net received energy; negative W_X is net supplied energy. The integral fixes the transfer, while the rest of line 4 remains open.

Take $L = 1 \text{ H}$, $R = 1 \Omega$, $C = 1 \text{ F}$ and $X = 1 \text{ H s}$. Put $u = t/(1 \text{ s})$ and obtain $f(t)$ by substituting each chosen trajectory into (3).

Supplier. For

$$q = (1 \text{ C}) \sin u, \quad i = (1 \text{ A}) \cos u, \quad q''' = -(1 \text{ A/s}^2) \cos u, \quad (7)$$

the inductive and capacitive terms cancel, as do the resistive and X terms, giving $f(t) = 0$. Hence

$$P_X = -(1 \text{ W}) \cos^2 u, \quad W_X(0, 2\pi \text{ s}) = -(1 \text{ J}) \int_0^{2\pi} \cos^2 u du = -\pi \text{ J}. \quad (8)$$

At $t = 0$, the component supplies 1 W; over a cycle it supplies $\pi \text{ J}$. Here $P_X = -Ri^2$ at every instant, so this transfer covers the resistor's absorption. The equation leaves its physical source unspecified.

Absorber. On $0 \leq t \leq 1 \text{ s}$, choose

$$q = (1 \text{ C}) \frac{u^3}{6}, \quad i = (1 \text{ A}) \frac{u^2}{2}, \quad q''' = 1 \text{ A/s}^2. \quad (9)$$

Now

$$f(t) = (1 \text{ V}) \left(1 + u + \frac{u^2}{2} + \frac{u^3}{6} \right), \quad P_X = (1 \text{ W}) \frac{u^2}{2}, \quad (10)$$

and

$$W_X = (1 \text{ J}) \int_0^1 \frac{u^2}{2} du = \frac{1}{6} \text{ J}. \quad (11)$$

At $t = 1 \text{ s}$, $P_{X \rightarrow 2} = -1/2 \text{ W}$. The same positive coefficient supplies energy on the sinusoid and absorbs it on this polynomial.

3 What choosing $X = LRC$ changes

For $(a, b, c) = (L, R, 1/C)$, *ODE Coefficient Synthesis*, Sections 1–3, selects

$$X = A_3 = A_{-1,3} = \frac{ab}{c} = LRC > 0. \quad (12)$$

Here $R > 0$. Since RC has units of time, $[X] = \text{H s}$ and Xq''' is a voltage. Equation (5) becomes $P_{X \rightarrow 2} = -LRC q''' i$.

Fix $q(t)$, including amplitude and time scale. Increasing LRC then strengthens whichever direction $q''' i$ already selects. The signed work scales the same way on a fixed interval. With independent positive bounds, the greatest magnitude uses all three upper bounds; without bounds there is no finite maximum. Recompute $f(t)$ to retain the chosen motion.

For $q = Q \sin(\omega t)$, with $Q, \omega > 0$,

$$q''' = -\omega^2 i, \quad P_{X \rightarrow 2} = LRC \omega^2 i^2 \geq 0. \quad (13)$$

All positive choices make X a supplier except at current zeros. Absorption needs a different derivative sign, as in (9). Relative to resistor absorption, at $i \neq 0$,

$$\frac{|P_{X \rightarrow 2}|}{Ri^2} = LC \left| \frac{q'''}{i} \right|. \quad (14)$$

This ratio is $LC\omega^2$ on a sinusoid. Increasing R scales both powers equally; only L and C change their ratio on fixed motion.

Use Section 2's supplier at $t = 0$ and absorber at $t = 1$ s, recomputing the forcing for each row.

Table 1. Signed contributions $P_{X \rightarrow 2}$ on the same two trajectories.

(L, R, C) in (H, Ω , F)	X (H s)	Supplier (W)	Absorber (W)
(1, 1, 1)	1	+1	−1/2
(1, 2, 1)	2	+2	−1
(2, 2, 2)	8	+8	−4

With $1 \leq L/(1 \text{ H}), R/(1 \Omega), C/(1 \text{ F}) \leq 2$, the last row maximizes both magnitudes. The same integrals give 8π J supplied per sinusoidal cycle and $4/3$ J absorbed over the polynomial interval.

If forcing and initial conditions are fixed instead, changing the coefficients changes the motion too. Maximizing LRC alone then says nothing about the maximum transfer.

4 Hammer and nail

Let x be hammer-face and nail-head movement during smooth contact, $v = x'$, and $m, d, k > 0$ mass, effective resistance and stiffness. The *domain map*, Sections 2–3, replaces (L, R, C) by $(m, d, 1/k)$:

$$X_h x''' + mx'' + dx' + kx = F(t), \quad X_h = \frac{md}{k}. \quad (15)$$

Assume the unknown component acts inside the hammer. Its received power is $X_h x''' v$; Section 3's fixed-motion scaling becomes md/k .

Think of m as the hammer's heft, d as the resistance encountered by the moving nail, and k as the stiffness of the contact, workpiece and backing. Here dv represents resisting losses by a simple speed-dependent force, while kx is the restoring force from elastic give. The unfamiliar x''' measures how quickly acceleration changes. While the head moves forward, a deceleration that builds in strength gives $x'''v < 0$, so the unknown component supplies power; a deceleration that eases gives the opposite sign, so it absorbs power.

For $F = 0$, $x(0) = 0$, $v(0) = v_0 > 0$ and $x''(0) = 0$, put $\omega = \sqrt{k/m}$. Up to the first stop,

$$x(t) = \frac{v_0}{\omega} \sin(\omega t), \quad 0 \leq t \leq t_* := \frac{\pi}{2\omega}. \quad (16)$$

Indeed, $mx'' + kx = 0$ and $X_h x''' + dv = 0$, so

$$W_{X_h} = \int_0^{t_*} X_h x''' v dt = - \int_0^{t_*} dv^2 dt = - \frac{\pi dv_0^2}{4\omega} = -W_d. \quad (17)$$

The component supplies the resistance work from its unspecified account.

To read this ideal blow in familiar terms, change one coefficient at a time, keeping the other two and the incoming speed v_0 fixed. The head travels $x(t_*) = v_0/\omega$ before its first stop.

Increasing m corresponds to using a heavier hammer. At the same head speed, there is more inertia to arrest: the blow takes longer to stop and the head travels farther. Reducing m gives a shorter movement and an earlier stop. The larger mass also raises X_h ; on these blows the unknown component supplies more work as resistance acts over the longer movement. Keeping the incoming speed the same does not mean that the heavier hammer takes the same effort to swing.

Increasing d represents stronger opposition at the same speed—the drag one associates with a nail that grips harder. Decreasing d represents easier movement. Extra drag would change an ordinary hammer's motion. In this particular blow, however, increasing d also increases X_h , so that the unknown component supplies exactly the extra resistance work. The stopping time and head travel stay unchanged, while the resistance work and the component's supplied work both increase in proportion to d . Reducing d reduces both. This cancellation is the distinctive prediction of the assumed component on the stated trajectory.

Increasing k corresponds to firmer backing: a board supported close to the nail gives less than one that can bend under the blow. At a given displacement the restoring force is larger, and this ideal blow stops sooner after less head travel. Decreasing k lets the movement continue longer and farther. Since compliance is $1/k$, firmer backing reduces X_h and, on these blows, the work supplied by the unknown component; softer backing increases both. Extra head travel can be elastic bending of the support. Permanent nail depth remains undetermined by this model.

For a coefficient of performance (COP), count work delivered to the modeled resistance and spring as output. Count initial kinetic energy and applied work as input, excluding the unknown component's supplied work. Call this ratio an apparent COP. For the stated blow, the endpoint energies and load work are

$$\begin{aligned} K_0 &= \frac{1}{2}mv_0^2, & U_* &:= \frac{1}{2}kx(t_*)^2 = K_0, \\ W_{\text{load},h} &= \int_0^{t_*} (dv + kx)v dt = W_d + U_*. \end{aligned} \quad (18)$$

Since $F = 0$, the applied work $\int_0^{t^*} F v dt$ is zero. Thus

$$\text{COP}_{\text{app},h} = \frac{W_{\text{load},h}}{K_0} = 1 + \frac{\pi d}{2\sqrt{mk}}. \quad (19)$$

For example, $d = \sqrt{mk}$ gives $1 + \pi/2 \approx 2.57$. Under the same one-at-a-time comparisons at fixed v_0 , increasing d raises the ratio, while increasing m or k lowers it. A heavier hammer receives more work from the component on this blow, but its initial kinetic energy grows faster. The ratio counts both dissipated work and recoverable elastic energy; neither quantity by itself measures permanent nail penetration.

5 Rotary output on prescribed motion

For output angle θ and speed $\Omega = \theta'$, substitute $(q, L, R, C, f) \mapsto (\theta, J, d, 1/k, \tau_h)$ in (3). Here $J, d, k > 0$ are output inertia, damping and torsional stiffness; τ_h is prescribed hammer torque and J excludes the hammer. The assumed component inside the tool has $X_r = Jd/k$. On identical motion, let τ_X and τ_0 be the required torques with and without it. Their input-work difference is

$$\Delta W_h = \int_{t_0}^{t_1} (\tau_X - \tau_0) \Omega dt = \int_{t_0}^{t_1} X_r \theta''' \Omega dt = W_{X_r}. \quad (20)$$

Supplied work therefore reduces the required hammer input on that motion.

For $\theta = \Theta \sin(\omega t)$, $\Theta, \omega > 0$, and $T = \pi/(2\omega)$, the relation $\theta''' = -\omega^2 \Omega$ gives

$$\begin{aligned} W_{X_r} &= \int_0^T X_r \theta''' \Omega dt = -\frac{\pi X_r \Theta^2 \omega^3}{4}, \\ W_d &= \int_0^T d \Omega^2 dt = \frac{\pi d \Theta^2 \omega}{4}, \quad \frac{-W_{X_r}}{W_d} = \frac{J \omega^2}{k}. \end{aligned} \quad (21)$$

When $J \omega^2/k \leq 1$, an ordinary model gives the same torque on this motion with nonnegative damping

$$d_{\text{eff}} = d - X_r \omega^2 = d \left(1 - \frac{J \omega^2}{k} \right). \quad (22)$$

For example, $J \omega^2/k = 1/10$ gives $d_{\text{eff}} = 9d/10$ and $W_{X_r} = -W_d/10$. The component supplies work that reduced damping avoids dissipating. This comparison adjusts the forcing to match motion; hammer preparation and the unknown account remain outside it.

Apply the same apparent-COP definition to the quarter-sine interval, taking $0 < r := J \omega^2/k \leq 1$. Its initial kinetic and final elastic energies are

$$U := \frac{1}{2} k \Theta^2, \quad K_0 = \frac{1}{2} J \Theta^2 \omega^2 = rU. \quad (23)$$

The corresponding load work and applied hammer work follow from their signed power integrals:

$$\begin{aligned} W_{\text{load},r} &= \int_0^T (d \Omega + k \theta) \Omega dt = U + W_d, \\ W_h &= \int_0^T \tau_X \Omega dt = U - K_0 + (1 - r) W_d = (1 - r)(U + W_d). \end{aligned} \quad (24)$$

Here $\tau_X = J\theta'' + d\Omega + k\theta + X_r\theta'''$ is the required hammer torque with the component. Including the output's initial kinetic energy in the denominator gives

$$\text{COP}_{\text{app},r} = \frac{W_{\text{load},r}}{K_0 + W_h} = \frac{U + W_d}{U + (1-r)W_d}. \quad (25)$$

For $r = 1/10$, additionally choosing $W_d = U$, equivalently $d\omega/k = 2/\pi$, gives $\text{COP}_{\text{app},r} = 20/19 \approx 1.053$. Supplying one tenth of the damping work therefore does not imply a ten-percent reduction in the total counted input.

In both examples, the excess over one comes from excluding the component's supplied work from the input. Including $-W_{X_h} = W_d$ or $-W_{X_r} = rW_d$ in the respective denominator makes the ratio exactly one. These interval identities leave the component's remaining account open. A practical tool COP additionally needs a defined useful nail-driving or fastening output and the energy cost of preparation and replenishment.

References

1. H. Nilre and B. C. Herlin (2026). [The ODE Template and Its Domain Equivalents](#).
2. H. Nilre and B. C. Herlin (2026). [ODE Coefficient Synthesis](#).
3. H. Nilre and B. C. Herlin (2026). [Interconnecting Series and Parallel ODEs](#).